

Notes: Darmouni and Mota (RFS, 2024)

The Savings of Corporate Giants

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1 Big picture

- **Question.** How do the largest U.S. nonfinancial corporations manage their financial assets, and what explains the rise of their large financial portfolios?
- **Setting.** The paper builds a hand-collected panel of financial-asset holdings for 200 large U.S. public firms from 2000 to 2021, with quarterly data for 2020 to zoom in on the COVID liquidity shock.
- **Core challenge.** Standard databases like Compustat do not properly separate cash from marketable securities and do not record the detailed asset composition that matters for understanding motives.
- **Main conceptual contribution.** The paper argues that looking only at the *level* of “cash” is incomplete. What matters is the *composition* of corporate financial assets: cash-like instruments, Treasuries, corporate bonds, equity, and other marketable securities can serve different purposes.
- **Main descriptive result.** Since 2007, total financial assets in the sample grew by about \$1 trillion, but only about one-third of that rise came from cash-like instruments. The majority of the growth came from marketable securities, especially corporate bonds.
- **Key portfolio fact.** By 2017, bonds were at least as large as cash-like instruments: cash-like assets were about 44% of financial assets, while U.S. government debt plus corporate bonds were about 45%.
- **Main mechanism result.** The 2017 Tax Cuts and Jobs Act triggers a sharp rundown in marketable securities, especially for tech, pharma, and high-foreign-income firms, while cash-like instruments remain roughly unchanged. This points to cross-border tax incentives as a main driver of marketable-securities accumulation.
- **Liquidity result.** During the COVID-19 liquidity shock, firms more exposed to the revenue collapse sharply increase cash-like instruments, but not corporate bond holdings. This supports the view that cash-like instruments serve precautionary liquidity motives.
- **Risk result.** Because many firms replaced cash-like assets with bond portfolios, the rise in marketable securities exposed them to duration risk. The paper shows that some corporate giants experienced historically large losses on their financial portfolios in 2022.
- **Economic takeaway.** “Cash” on large corporate balance sheets is not one object. Marketable securities seem primarily linked to tax optimization, while cash-like instruments are the assets firms rely on when they truly need liquidity.

2 Data and measurement (key objects)

2.1 Why this paper needs new data

- Compustat records CH (cash) and CHE (cash and short-term investments), but it does not accurately separate cash from marketable securities.
- It also does not record the share of financial assets reported in noncurrent assets.

- The footnotes of 10-K annual reports, by contrast, often report detailed fair-value tables with separate line items for cash, money market funds, Treasuries, corporate securities, equities, and other asset classes.

Interpretation. The empirical innovation of the paper is simple but powerful: instead of relying on balance-sheet aggregates, it manually reads the firms’ financial instrument tables and reconstructs the portfolio composition directly.

2.2 Sample construction

- The sample covers 200 large U.S. public firms from 2000 to 2021.
- The starting point is the union of the 100 largest Compustat firms by total assets in 2000, 2009, and 2017, which yields 166 firms.
- The authors then add 44 more large firms from 2017 that were not already covered.
- The sample includes only firms listed on NYSE, Nasdaq, or Amex.
- Financial firms, regulated utilities, and public-service-type entities are excluded.
- For 2020, the data are quarterly to capture the COVID shock at higher frequency.

In recent years, the sample captures about 63% of aggregate Compustat **cash and short-term investments** among publicly traded firms.

Interpretation. The paper is not trying to represent every U.S. corporation. It is deliberately about *corporate giants*, the firms that dominate aggregate financial-asset holdings.

2.3 How financial assets are defined

Let total financial assets for firm i at time t be

$$FA_{it} = CashLike_{it} + MarketableSec_{it}. \quad (1)$$

The paper’s baseline classification is:

- **Cash-like instruments:** cash, money market funds, deposits, and commercial paper when separately reported; otherwise total “cash and cash equivalents.”
- **Marketable securities:** all non-cash-like financial assets.

A further decomposition is

$$MarketableSec_{it} = USGov_{it} + CorpBond_{it} + Equity_{it} + Other_{it}, \quad (2)$$

where U.S. government includes Treasuries and agency debt.

Interpretation. The key conceptual line is not “safe” versus “risky” in the abstract; it is *cash-like liquidity management* versus *non-cash-like portfolio investment*. Treasuries sit in the middle: they are safe securities, but they are still marketable securities rather than pure cash buffers.

2.4 Measurement caveats

- Disclosure of outstanding values is required, but consistent disclosure of maturity, yield, or duration is not.
- The paper therefore cannot build a full security-level risk map for every firm-year.
- Breakdown of cash-like subcomponents is missing especially before 2009, so the paper avoids strong claims about the internal composition of the cash-like bucket.
- When detailed breakdown is unavailable, all “cash and cash equivalents” are conservatively assigned to the cash-like category.

Interpretation. The paper is strongest on the broad split between cash-like instruments and marketable securities, and weaker on the exact maturity structure within each asset class.

2.5 Main variables used in the event studies

For the tax-reform and COVID analyses, the outcome variables are typically scaled by book assets:

$$\frac{FA_{it}}{AT_{it}}, \quad \frac{CashLike_{it}}{AT_{it}}, \quad \frac{MarketableSec_{it}}{AT_{it}}, \quad \frac{USGov_{it}}{AT_{it}}, \quad \frac{CorpBond_{it}}{AT_{it}}. \quad (3)$$

The payout variable is

$$\frac{Dividends_{it} + NetRepurchases_{it}}{AT_{i,t-1}}. \quad (4)$$

For the 2022 risk analysis, the estimated financial portfolio return is

$$r_{it}^p = \frac{FinancialIncome_{it}}{FA_{i,t-1}}, \quad (5)$$

where financial income is the sum of interest income, dividend income, and net realized and unrealized gains on financial assets.

Interpretation. Scaling by assets makes the cross-firm comparisons interpretable and avoids confusing large firms simply being large with firms choosing systematically different portfolio policies.

2.6 Descriptive facts that matter

Table 1 summarizes the big asset classes.

- Total financial assets rise from \$325.5B in 2000 to \$1.13T in 2012 and \$1.68T in 2017, before falling to \$1.53T in 2020.
- Cash-like instruments rise from \$160.9B in 2000 to \$739.2B in 2017, but their *share* of financial assets falls from 49.45% in 2000 to 43.91% in 2017.
- Corporate bonds rise from only \$29.2B in 2000 to \$404.8B in 2017, and their share rises from 8.96% to 24.05%.
- U.S. government securities rise from \$26.2B in 2000 to \$349.5B in 2017 and account for roughly 20% of financial assets in both 2012 and 2017.

The paper also emphasizes concentration:

- the top 10 firms hold over half of total financial assets in 2017,
- the top 20 firms hold about 76%,
- and the top 10 firms hold over 70% of aggregate marketable securities.

Apple alone held about 40% of all corporate bond positions in the sample in 2017.

Interpretation. The aggregate facts are really giant-firm facts. This is why the paper focuses on the very top of the firm-size distribution rather than on the median corporation.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and hypotheses

The paper studies two broad motives for financial-asset accumulation.

- **Liquidity motive.** Firms hold safe, liquid assets to self-insure against bad shocks or financing frictions.
- **Cross-border tax motive.** Multinationals keep earnings abroad and park them in financial assets instead of repatriating and distributing them.

The portfolio-composition predictions differ.

- If assets are held mainly for liquidity, firms should want cash-like instruments in hard times.
- If assets are held mainly for tax optimization, firms can hold yield-bearing marketable securities and unwind them when tax incentives change.

Interpretation. The paper’s whole empirical design is built around the idea that *different assets should react differently to different shocks*. That is what lets composition identify motive.

3.2 TCJA event-study / difference-in-differences specification

To study the 2017 tax reform, the paper estimates

$$y_{it} = \beta_1 (Post_t \times \mathbf{1}\{Tech_i\}) + \beta_2 (Post_t \times \mathbf{1}\{Pharma_i\}) + \alpha_i + \alpha_t + \varepsilon_{it}, \quad (6)$$

where:

- y_{it} is an asset-share or payout outcome,
- $Post_t = 1$ for 2018–2019 and 0 for 2012–2017,
- α_i are firm fixed effects,
- α_t are year fixed effects.

The paper also estimates a related variant:

$$y_{it} = \beta(Post_t \times HighForeignIncome_i) + \alpha_i + \alpha_t + \varepsilon_{it}. \quad (7)$$

Interpretation. This is not a firm-level treatment effect in the experimental sense. It is a differential-response design: firms more exposed to multinational tax incentives should react more strongly after the TCJA if tax motives are central.

3.3 What Table 2 says numerically

In the tech-versus-other comparison, after the TCJA:

- total financial assets / assets falls by 6.88 percentage points for tech firms,
- marketable securities / assets falls by 6.15 percentage points,
- cash-like / assets changes by only -0.73 percentage points and is not statistically strong,
- payouts / lagged assets rise by 4.74 percentage points.

For pharma firms, the pattern is similar: total financial assets fall by 4.88 percentage points, marketable securities fall by 4.94 percentage points, and payouts rise by 1.89 percentage points.

Interpretation. The main post-TCJA action is not “firms stopped hoarding cash.” It is “firms unwound securities portfolios and paid out cash to shareholders.” That is much more consistent with tax-driven marketable-security accumulation than with pure precautionary savings.

3.4 COVID liquidity-shock specification

To study the 2020 crisis, the paper estimates

$$y_{it} = \gamma(Post_t \times CovidExposure_i) + \alpha_i + \alpha_t + \varepsilon_{it}, \quad (8)$$

where:

- $Post_t$ marks the post-COVID quarters beginning in 2020Q1,
- $CovidExposure_i$ is an industry-level measure based on the abnormal employment decline in the firm’s three-digit NAICS industry between 2019Q2 and 2020Q2 relative to its prior five-year trend,
- α_i are firm fixed effects,
- α_t are quarter fixed effects.

Interpretation. The identifying idea is that industries hit harder by the COVID revenue shock should have stronger precautionary demand for liquid assets.

3.5 What Table 3 says numerically

A one-standard-deviation increase in COVID exposure is associated with:

- a 1.9 percentage point increase in total financial assets over assets,
- a 1.3 percentage point increase in cash-like instruments over assets,
- a 0.6 percentage point increase in marketable securities over assets,
- a 0.3 percentage point increase in U.S. government securities over assets,
- and no statistically meaningful increase in corporate bond holdings.

Interpretation. When firms urgently need liquidity, they move first into cash-like instruments, not into corporate bonds. This is the cleanest evidence in the paper that cash-like assets and marketable securities serve different functions.

3.6 Portfolio-return measurement for the 2022 risk exercise

For selected large firms, the paper estimates

$$Return_{it}^{portfolio} = \frac{Interest_{it} + Dividends_{it} + RealizedGains_{it} + UnrealizedGains_{it}}{FA_{i,t-1}}. \quad (9)$$

It also computes the spread over the one-month Treasury bill:

$$Spread_{it} = Return_{it}^{portfolio} - r_t^{Tbill}. \quad (10)$$

Interpretation. This is not a full mark-to-market portfolio accounting system for all firms. It is a carefully assembled partial return measure for five major firms, used to show that duration risk became economically important in 2022.

3.7 Panel structure and estimation choice

- The TCJA design uses annual data from 2012 to 2019 with firm and year fixed effects.
- The COVID design uses quarterly data around 2018Q4 to 2021Q4 with firm and quarter fixed effects.
- The paper does not rely on a structural asset-pricing model or a full dynamic corporate-finance model; it uses transparent event-study style difference-in-differences regressions.

Interpretation. The empirical strategy is deliberately simple. The contribution comes more from the new portfolio-composition data than from econometric complexity.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the TCJA estimate

The tax-reform estimates come from comparing firms that should be more exposed to cross-border tax incentives with firms that should be less exposed, before and after the TCJA.

Why tech and pharma? Because these sectors rely more on intangible assets, making it easier to book income abroad and accumulate offshore financial assets.

Why high foreign income? Because foreign income is a more direct proxy for tax-exposed multinationals.

Interpretation. The identification is not “policy changed for some firms but not others.” It is “policy changed for everyone, but firms with stronger ex ante incentives to keep funds abroad should react more.”

4.2 Why reverse causality is relatively weak here

Could lower firm performance cause both portfolio runoff and higher payouts? In a mechanical sense, yes, but that is not the pattern observed.

- The TCJA is a common national shock with a sharp date.
- The portfolio adjustment is concentrated in securities rather than in cash-like balances.
- The strongest effects are in the sectors where offshore profit shifting is most plausible.

Interpretation. If the story were simply “some firms needed cash,” we would expect a broader drawdown of liquidity balances. The fact that firms sell bonds rather than run down cash-like assets is precisely why the tax interpretation is persuasive.

4.3 What identifies the COVID estimate

The COVID design exploits heterogeneous exposure to the sudden revenue shock using an external industry measure based on abnormal employment losses.

This helps because firms do not choose their 2020 industry shock in response to their portfolio policy, and the shock is tightly linked to precautionary liquidity demand.

Interpretation. The COVID event study is a classic demand-for-liquidity test. Firms more exposed to the collapse in cash flow respond by accumulating the most liquid assets.

4.4 Why composition is more informative than levels

A rise in total financial assets alone cannot distinguish among:

- taxes,
- financing frictions,
- intangible intensity,
- governance,
- or a general taste for saving.

But the paper’s composition evidence is much sharper:

- the TCJA mostly reduces marketable securities,

- COVID mostly raises cash-like instruments,
- 2022 losses show that securities portfolios carry real duration risk.

Interpretation. The composition shifts are the core identification argument of the paper.

4.5 Alternative explanations the paper discusses

The paper is careful not to overclaim. It discusses several nonexclusive possibilities.

- Some unconstrained firms may rationally prefer higher-yield but less liquid securities.
- Managerial misperception or WACC-style mistakes could also make higher-yield assets look attractive.
- A distinct hypothesis is that firms tilted toward illiquid assets because the TCJA taxed noncash assets at a lower rate than cash-equivalent assets.

The paper argues that the last explanation is hard to fully support because implied repatriation tax rates for the biggest holders do not line up neatly with their marketable-securities shares.

Interpretation. The evidence is strongest for the broad tax-versus-liquidity distinction, not for the exact micro-mechanism inside the tax motive.

4.6 What the paper can and cannot distinguish

- It can distinguish the roles of *cash-like* assets and *marketable securities*.
- It can show that different macro events move these two buckets differently.
- It cannot directly trace where inside a multinational structure the assets are held.
- It cannot observe security-level duration, credit risk, or yields consistently for the full sample.
- It cannot fully separate tax optimization from every other nonliquidity motive for holding securities.

Interpretation. This is a data paper with strong event-study evidence, not a fully closed causal decomposition of every corporate-saving motive.

5 Tables and figures

5.1 Figure 1 and Figure 2: what these firms actually hold

Read:

- Figure 1 shows the 2017 aggregate composition: 44% cash-like, 24% corporate bonds, 20.7% U.S. government securities, 2.03% equity, and 9.24% other assets.
- Figure 2 shows Apple's 2017 fair-value table and makes the paper's point tangible: Apple held roughly \$268.9B in financial assets, including \$152.7B of corporate securities and \$55.3B of U.S. Treasury securities.

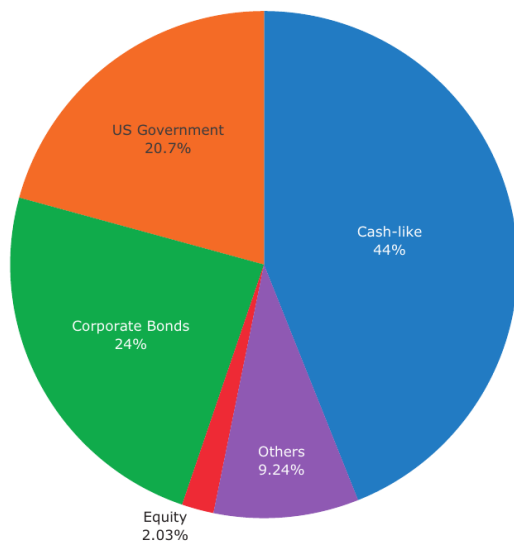


Figure 1
Financial assets portfolio composition of nonfinancial corporations in 2017
This figure displays the financial assets portfolio composition in 2017. "Cash-like" is defined as the sum of cash, MMF, deposits, and commercial paper if this information is reported, and "cash and cash equivalents" otherwise. "U.S. government" bonds include Treasuries and Agency securities.

2017				
	Adjusted Cost	Unrealized Gains	Unrealized Losses	Fair Value
Cash	\$ 7,982	\$ —	\$ —	\$ 7,982
Level 1:				
Money market funds	6,534	—	—	6,534
Mutual funds	799	—	(88)	711
Subtotal	7,333	—	(88)	7,245
Level 2:				
U.S. Treasury securities	55,254	58	(230)	55,082
U.S. agency securities	5,162	2	(9)	5,155
Non-U.S. government securities	7,827	210	(37)	8,000
Certificates of deposit and time deposits	5,832	—	—	5,832
Commercial paper	3,840	—	—	3,840
Corporate securities	152,724	969	(242)	153,451
Municipal securities	961	4	(1)	964
Mortgage- and asset-backed securities	21,684	35	(175)	21,544
Subtotal	253,084	1,278	(694)	253,668
Total	\$ 268,399	\$ 1,278	\$ (782)	\$ 268,895

Figure 2
Apple financial instruments table
This figure displays the Financial Instrument Table for Apple in 2017. Available SEC Edgar.

saving U.S. taxes.³ The Tax Cuts and Jobs Act (TCJA) aimed to reduce

Economic message. These are the exhibits that break the standard “corporate cash” narrative. The giant firms in this paper are not simply sitting on bank deposits; they are actively managing large bond portfolios.

5.2 Figure 3, Figure 4, and Table 1: growth, composition, and concentration

Read:

- Figure 3 shows aggregate financial assets rising strongly up to 2017 and then partially reversing.
- Table 1 shows that corporate bonds become the single largest noncash asset class by 2017.
- Figure 4 shows that the aggregate rise is heavily concentrated in tech and pharma: these sectors have both higher financial-assets-to-assets ratios and faster asset growth.

Economic message. The rise in financial assets is not a broad-based firm phenomenon. It is driven by giant firms, especially in intangible-intensive sectors, and within those firms by a shift toward marketable securities.

5.3 Figure 5 and Table 2: the TCJA reversal

Read:

- Figure 5 shows the sharp break around the tax reform: total financial assets fall, corporate bonds drop sharply, and cash-like instruments remain much more stable.

2. The Growth in Financial Assets of Corporate Giants

2.1 Large aggregate growth until 2017

Figure 3 displays the growth in financial assets in our sample. The fast growth is in line with existing evidence on aggregate “cash” holdings and the rise in corporate savings (Chen, Karabournis, and Neiman 2017). Aggregate

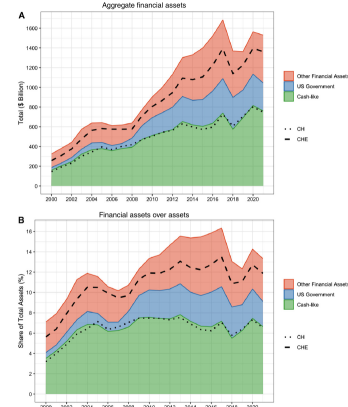


Figure 3
The growth in financial assets
This figure plots the growth in financial assets for our sample of firms. Panel A plots aggregate financial assets, while panel B plots total financial assets over total assets. “Cash-like” is defined as the sum of cash, MMF, deposits, and commercial paper if this information is reported, and “cash and cash equivalents” otherwise. “U.S. Government” includes Treasuries and Agency securities. “CH” is Cash and “CHB” is Cash and Short-Term investment from Compustat.



Figure 4
Sectoral composition effects behind aggregate growth
These figures plot the aggregate growth for three sectors in our sample: pharmaceutical firms, technological firms, and others, as defined in Table A.1 in the Internet Appendix. The left panels present the growth in the ratio of average financial assets over average assets within each industry. The right panels show the growth of aggregate total assets within each industry relative to 2010. The top panel uses book values of assets (book value of equity + total liabilities), while the bottom panel uses market values (market value of equity + total liabilities).

balances. Table 1 shows that when U.S. government debt and corporate bonds are combined, they represent 45% of financial assets in 2017, whereas cash-like instruments account for 44%. In 2012, the proportions were inverted, with 33% in bonds versus 50% in cash-like instruments. The share of cash-like instruments to financial assets has fallen five percentage points between 2000 and 2017.

Moreover, we document the meteoric rise of corporate bonds in the aggregate portfolio. Corporate bond holdings have almost tripled in value between 2012 and 2017 to reach \$400 billion. They constitute almost 25% of aggregate financial assets, which makes them the single largest asset class, according to our classification. In particular, they have outgrown U.S. Treasuries and agencies whose share has been stable since 2012 at around 20%.

This is surprising, as other corporations issue these bonds. In contrast with the textbook view that economic firms only demand outside financial investments

Table 1
Composition of financial assets

	Total in USD Bn				Share of Financial Assets (%)				Share of Total Assets (%)			
	2000	2012	2017	2020	2000	2012	2017	2020	2000	2012	2017	2020
Cash-like instruments	891.01	971.04	776.22	762.13	48.45	50.46	43.94	40.81	3.31	7.40	11.86	6.64
U.S. government	26.22	227.01	349.49	283.60	1.45	20.07	20.76	16.50	0.37	2.94	3.39	2.47
Corporate bonds	29.18	144.79	404.79	261.88	1.56	12.80	24.05	17.11	0.64	1.88	5.93	2.28
Equities	28.75	274.01	34.28	67.24	1.53	2.42	2.04	4.39	0.63	6.33	0.33	0.59
Others	80.42	100.63	155.62	153.65	4.31	24.71	14.20	9.24	10.19	1.76	2.08	1.51
Total	1,225.58	1,728.09	1,681.41	1,550.09	100	100	100	100	100	100	100	100

This table displays the composition of financial assets in our sample. “Cash-like” is defined as the sum of cash, MMF, deposits, and commercial paper if this information is reported, and “cash and cash equivalents” otherwise. “U.S. government” includes Treasuries and agency debt. “Others” contain all items that are clearly not cash-like instruments, are not U.S. government securities, are not corporate debt or equities, or are difficult to classify due to ambiguous language (e.g., “other securities”).

sample period due to its massive holdings of corporate bonds and low

- Table 2 shows that the post-TCJA contraction is strongest for tech and pharma firms and is concentrated in marketable securities.
- For tech firms, marketable securities fall by 6.15 percentage points of assets and payouts rise by 4.74 percentage points of lagged assets.

Economic message. This is the paper’s cleanest evidence that the buildup of securities portfolios was linked to offshore tax incentives. The reform did not mainly unwind operating cash buffers; it unwound large securities portfolios.

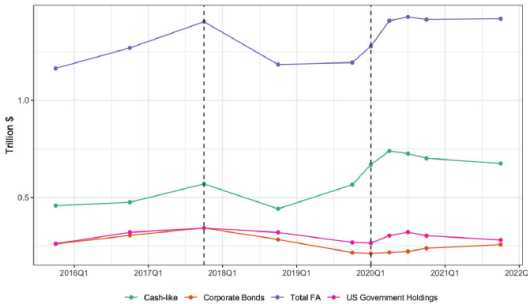


Figure 5
Aggregate portfolio dynamics, 2015–2021

This figure plots the aggregate dynamics of different asset classes from 2015 to 2020Q1 in our sample. The first vertical dashed line corresponds to the TCJA. The second vertical dashed line corresponds to the COVID-19 liquidity crisis. “Cash-like” is defined as the sum of cash, MMF, deposits, and commercial paper if this information is reported, and “cash and cash equivalents” otherwise. “U.S. government holdings” include Treasuries and agency debt.

Table 2
Effects of tax reform: Difference-in-differences regressions

A. Tech versus pharma						
	Total FA / AT (%) (1)	Cash-like / AT (%) (2)	Marketable securities/AT (%) (3)	U.S. government bonds/AT (%) (4)	Corporate bonds/AT (%) (5)	Payouts / lagged AT (%) (6)
Post:tech	−6.881*** (0.949)	−0.730 (0.736)	−6.151*** (0.730)	−2.135*** (0.337)	−1.795*** (0.451)	4.740*** (0.938)
Post:pharma	−4.878*** (1.253)	0.059 (0.971)	−4.937*** (0.964)	−1.118** (0.444)	−2.051*** (0.595)	1.889 (1.238)
Observations	1,331	1,331	1,331	1,331	1,331	1,331
R ²	.863	.616	.880	.928	.785	.561
Adjusted R ²	.841	.555	.861	.916	.751	.491
B. Foreign income share						
	Total FA / AT (%) (1)	Cash-like / AT (%) (2)	Marketable Securities/AT (%) (3)	U.S. government Bonds/AT (%) (4)	Corporate Bonds/AT (%) (5)	Payouts / Lagged AT (%) (6)
Post:high_foreign_income	−2.802*** (0.790)	1.094* (0.600)	−3.896*** (0.607)	−1.212*** (0.277)	−1.317*** (0.370)	1.509* (0.773)
Observations	1,331	1,331	1,331	1,331	1,331	1,331
R ²	.857	.617	.876	.926	.783	.553
Adjusted R ²	.834	.556	.856	.914	.749	.482

This table displays estimated coefficients of difference-in-differences regression of the components of financial assets shares on sector dummies with year and firm fixed effects. “Post” is a dummy variable, defined as one for year 2018 to 2019, and as zero for year 2012 to 2017. Only data from 2012 to 2019 in our sample are included in the regressions. “Tech” and “Pharma” are dummy variables defined as one for technological and pharmaceutical firms, respectively. “High_foreign_income” is a dummy variable defined as one if the average foreign income as a share of total sales between 2012 and 2016 is above median and zero otherwise. Details about the classification of industries could be found in Table 5. “Cash-like” is defined as the sum of cash, MMF, deposits, and commercial paper if this information is reported, and “cash and cash equivalents” otherwise. “Marketable securities” are non-cash-like financial assets. “U.S. government” includes Treasuries and Agency debt. All columns include firm and year fixed effects. Column 6 examines payouts (dividends + net equity repurchases) over lagged assets. Standard errors are in parentheses.

5.4 Table 3: the COVID dash for cash

Read:

- Table 3 shows that firms in industries more exposed to the COVID cash-flow shock increased financial assets, but mainly through cash-like instruments.
- The cash-like response (1.3 percentage points) is much larger than the U.S. government-securities response (0.3 percentage points), and corporate bonds do not respond significantly.

Economic message. This is the key exhibit for the liquidity interpretation. When firms suddenly need buffers, they choose cash-like instruments first.

Table 3
Effects of COVID-19 shock: Difference-in-differences regression

	Dependent variable:				
	Total FA / AT (%)	Cash-like / AT (%)	Marketable securities/AT (%)	U.S. government bonds/AT (%)	Corporate bonds/AT (%)
	(1)	(2)	(3)	(4)	(5)
Post Covid exposure	0.019*** (0.005)	0.013*** (0.004)	0.006** (0.003)	0.003** (0.001)	0.003 (0.002)
Observations	720	720	720	720	720
R ²	.018	.057	.045	.079	.058
Adjusted R ²	.001	.006	.034	.074	.028

This table displays estimated coefficients of difference-in-differences regression of the components of financial assets shares on COVID-19 exposure with year and firm fixed effects. "Post" is a dummy variable, defined as one for quarters 2020Q1 to 2021Q4, as zero for 2018Q1 to 2019Q4. Only quarterly data from 2018Q4 to 2021Q4 in our sample are included in the regression. "Covid exposure" is the abnormal change in industry employment as in [Chokwe-Ritch et al. \(2022\)](#), calculated as the negative of the percentage change in national employment in the firm's three-digit NAICS industry between 2019Q3 and 2020Q2 less the trailing 5-year change. "Cash-like" is defined as the sum of cash, MMF, deposits, and commercial paper if this information is reported, and "Cash and cash equivalents" otherwise. "Marketable securities" are non-cash-like financial assets. "U.S. government" includes Treasury and Agency debt. All columns include firm and quarter fixed effects. Standard errors are in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

hypothesis that firms most affected by the COVID-19 shock increased savings in liquid assets.

Column 1 shows the increase in financial assets relative to other industries, and columns 2 to 5 reveal that the increase is mostly driven by the accumulation of cash-like instruments. A one-standard-deviation increase in *Covid_Exp* leads to a 1.9-p.p. increase in financial assets over assets, which is a 16% increase with respect to the average in the non-financial *Cash-like* instruments

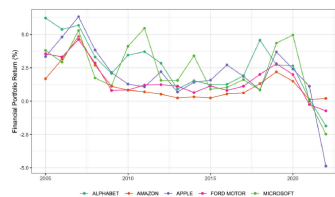


Figure 6
Estimated portfolio returns.

This figure plots the estimated annual portfolio returns dynamics from 2005 to 2022. Returns are defined as the ratio between financial income and the lag of total financial assets. Financial income is the sum of interest and dividends income, net realized gains, and net unrealized gains on marketable securities.

data on returns on firms' financial portfolios, we carefully investigate their financial statements to estimate financial income, defined as the sum of interest and dividends income and net realized and unrealized gains, divided by lagged financial assets.²⁷

Figure 6 shows the time series of estimated annual returns for the five selected firms. A firm like Apple experienced a negative 5% return on its

Table 4
Portfolio income analyses

Firm	A. The portfolio income, 2007–2021			
	Return (%)	Return spread over T-bill (%)	Income (M \$)	Cash-like share (%)
ALPHABET	2.53	1.76	1,360.71	27.49
MICROSOFT	2.55	1.59	1,976.87	7.85
APPLE	2.21	1.35	2,535.96	11.96
FORD MOTOR	1.53	0.75	636.73	41.39
AMAZON	1.19	0.41	201.47	58.36

Firm	B. The portfolio income, 2022			
	Return (%)	Return spread over T-bill (%)	Income (M \$)	Cash-like share (%)
APPLE	-4.87	-5.45	-9,279	16.19
MICROSOFT	-2.47	-2.60	-3,360	11.16
ALPHABET	-1.87	-3.39	-3,540	15.54
FORD MOTOR	-0.71	-2.15	-560	30.46
AMAZON	0.21	-1.22	166	38.72

This table displays the estimated financial portfolio, returns, excess returns over Treasury bill, total financial income, and the average cash-like share for Alphabet, Microsoft, Apple, Ford, and Amazon. Panel A uses data only from 2007 to 2021, and panel B uses only 2022 data. Financial income is the sum of interest and dividends income, net realized gains, and net unrealized gains on marketable securities. Return, r_{it} , is the financial income over the lagged financial assets, r_{it}/A_{it-1} , is the annually cumulative 1-month Treasury-bill return from CRSP computed over each firm's fiscal year, and return spread over Treasury bill is the average difference between the firm's r_{it} and the 1-month r_{T-bill} . The cash-like share is the average cash-like instrument as a share of total financial assets. "Cash-like" is defined as the sum of cash, MMF, deposits, and commercial paper if this information is reported, and "Cash and cash equivalents" otherwise. Interest and dividends income are derived from the table for other income. Net realized gains and net unrealized gains on marketable securities are obtained from the statement of other comprehensive income, when available, along with the information from the statements for shareholders' equity, from the table for other income, and from the footnotes of the annual reports.

5.5 Figure 6 and Table 4: the downside of large securities portfolios

Read:

- Figure 6 shows estimated portfolio returns for Apple, Microsoft, Alphabet, Ford, and Amazon from 2005 to 2022.
- Table 4 shows that 2007–2021 average portfolio returns were positive and larger for firms with lower cash-like shares.
- In 2022, however, Apple's estimated portfolio return is -4.87% and its financial-income loss is about \$9.3B; Microsoft and Alphabet also post large losses.

Economic message. The same shift toward yield-bearing securities that may have been attractive before 2022 also created meaningful duration risk once rates jumped. The corporate sector, not only banks, was exposed.

6 Short summary

- The paper builds new hand-collected data on the financial-asset portfolios of 200 large U.S. nonfinancial firms.
- The big descriptive result is that the rise in corporate financial assets up to 2017 was driven mainly by marketable securities, especially corporate bonds, not by cash alone.

- The TCJA of 2017 leads firms—especially tech, pharma, and high-foreign-income firms—to unwind marketable securities and raise payouts.
- The COVID shock of 2020 leads exposed firms to increase cash-like instruments, not corporate bond holdings.
- The 2022 rate shock reveals that the rise in securities portfolios had created nontrivial duration risk for some corporate giants.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that the financial assets of large U.S. nonfinancial firms are not well described as “cash.” These firms hold and actively rebalance large portfolios of marketable securities. The event-study evidence shows that marketable securities and cash-like instruments respond differently to tax and liquidity shocks, implying they serve different economic functions.

7.2 Economic intuition

Cross-border tax incentives make it attractive for multinationals to accumulate financial assets abroad and to park them in securities that earn yield. Liquidity management, by contrast, requires assets that remain safe and usable in bad states. This is why the TCJA mostly shrinks securities portfolios, while COVID mostly raises cash-like balances.

7.3 Takeaway

The paper’s main lesson is: for corporate giants, the composition of financial assets is as important as the level. Once we look inside the portfolio, the long-running debate about why firms hold so much “cash” becomes much easier to organize.